

	Method	Architecture	T	Accuracy
without KD	TET [11]	R-34	6	64.79%
	GLIF [55]	R-34	4	67.52%
	MPBN [18]	R-18	4	63.14%
		R-34	4	64.71%
	SEW ResNet [15]	R-18	4	63.18%
		R-34	4	67.04%
	Spikformer [64]	S-8-384	4	70.24%
		S-6-512	4	72.46%
with KD	Spikingformer [61]	Sg-8-384	4	72.45%
	Spike-driven Transformer [54]	Sg-8-384	4	72.28%
	Teacher ANN	SEW R-18	1	71.69%
		SEW R-34	1	75.38%
	LaSNN [24]	SEW R-18	4	63.33%
		SEW R-34	4	66.98%
	KDSNN [50]	SEW R-18	4	63.61%
		SEW R-34	4	67.28%
	BKDSNN [51]	SEW R-18	4	63.43%
		SEW R-34	4	67.21%
	Ours	SEW R-18	4	69.24% ± 0.19
		SEW R-34	4	73.16% ± 0.15

Table 2. Comparison results with training-based SNN SOTA methods, including CNN- and Transformer-based approaches, on ImageNet, with and without Knowledge Distillation (KD). **T** denotes Timestep. R, S, Sg denotes ResNet, Spikeformer and Spikingformer architectures respectively.

Temporal Separation		Entropy Regularization	Accuracy
KL Loss	CE Loss		
✗	✗	✗	77.15%
✗	✓	✗	77.41%
✓	✗	✗	77.64%
✓	✓	✗	77.99%
✓	✓	✓	78.13%

Table 3. Ablation of Temporal Separation Strategy and Entropy Regularization.

racy discrepancy among time steps for the vanilla SNN KD is 0.98%, notably higher than the 0.65% difference with our approach. Furthermore, the vanilla SNN KD yields an average output accuracy of 74.67%, whereas our method achieves 76.56%. And as the accuracy of individual time step outputs improved, we observe a significant enhancement in performance following the mean operation on these outputs, aligning with our hypothesis. This clearly demonstrates the effectiveness of our method and highlights the significance of temporal separation in the SNN distillation process.

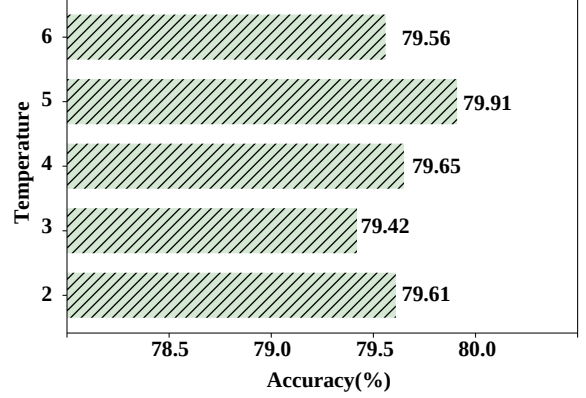


Figure 4. Temperature Coefficient Sensitivity. Our method demonstrates stability across various τ hyperparameters. This experiment is conducted on CIFAR100 with ResNet-34 as the teacher model and ResNet-18 as the student model.

Selection of Parameter λ . The entropy regularization term is essential for $\mathcal{L}_{TSER}$, as it partially alleviates the erroneous knowledge transferred by the teacher network. Thus, selecting an optimal value for λ is crucial. We conduct experiments with various values of λ , and the results are shown in Fig. 3. Increasing λ enhances the correction of the teacher’s inaccuracies but also dilutes the distribution of the correct knowledge. Our findings indicate that maintaining λ within the range of 0.0001 to 0.2 enhances the network’s overall training stability. However, excessively high values of λ can result in negative loss values, as described in Eq. 11, which must be avoided.

Temperature Coefficient Sensitivity. In our experiments, we configure the temperature to $\tau = 5$. Utilizing the prior experimental setup, we select five temperature points: $\tau = [2, 3, 4, 5, 6]$. We evaluate performance with ResNet-34 as the teacher model and ResNet-18 as the student model on the CIFAR100 dataset. The results depicted in Fig. 4, indicate that our method exhibits stable performance across different temperature coefficients.

4.4. Performance Analysis and Visualization

Firing Rate Analysis. In Table. 4, the firing rates of our proposed method show a slight increase compared to vanilla SNN KD across both ResNet and VGG architectures. Fig. 2 illustrates that an increase in firing rate of approximately 0.1% corresponds to a 1.48% improvement in overall performance. Notably, when comparing our method to other distillation techniques within the ResNet architecture, it maintains performance gains while simultaneously reducing firing rate and increasing spike sparsity, thus minimizing redundant spikes. Although LaSNN exhibits lower firing rates, this comes at the expense of some model per-

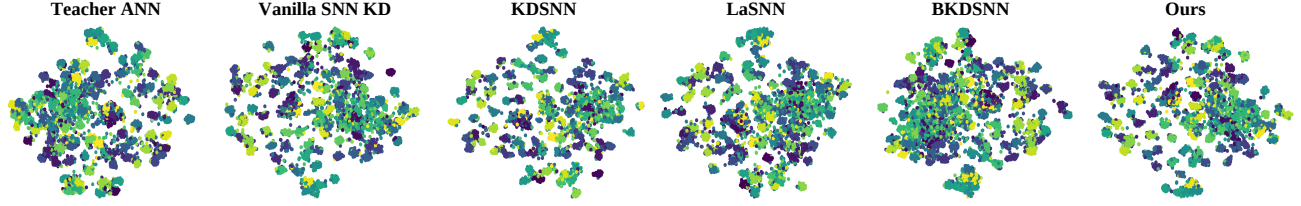


Figure 5. t-SNE Visualization of features learned by teacher ANN and different distillation methods.

Architecture	Methods	OPs	ACs	MACs	Energy	Training Time	Firing Rate
ResNet18	Vanilla SNN KD	2.23G	405.33M	12.2M	0.42mJ	2.78s	22.85%
	KDSNN	2.23G	411.49M	12.2M	0.43mJ	2.67s	23.23%
	LaSNN	2.23G	300.94M	12.2M	0.33mJ	2.79s	16.76%
	BKDSNN	2.23G	511.57M	12.2M	0.52mJ	2.84s	34.93%
	Ours	2.23G	405.98M	12.2M	0.42mJ	2.70s	22.95%
VGG16	Vanilla SNN KD	1.26G	159.96M	274.2M	1.40mJ	2.43s	18.91%
	KDSNN	1.26G	170.79M	274.2M	1.39mJ	2.39s	18.59%
	LaSNN	1.26G	150.96M	274.2M	1.41mJ	2.44s	17.25%
	BKDSNN	1.26G	171.94M	274.2M	1.40mJ	2.58s	18.66%
	Ours	1.26G	160.92M	274.2M	1.41mJ	2.40s	20.05%

Table 4. Energy consumption, training time, and spike firing rate under various distillation methods and architectures on the CIFAR100 dataset. Training Time refers to the time required per batch during training.

formance. Furthermore, unlike our logit-based approach, LaSNN employs a feature-based method that optimizes the feature map directly, making it more advantageous for feature sparsification.

Energy Consumption Analysis. Our energy consumption analysis, based on the energy model in [29], is summarized in Table. 4. The results reveal that the most significant variations in energy consumption occur in Accumulation Computation (AC), which is related to the firing rate. Notably, a lower firing rate correlates with reduced AC. Consequently, our approach does not increase the computational load of SNNs and is competitive regarding energy efficiency.

Training Time. We evaluate the training times of various competitive KD methods on the CIFAR100 dataset, focusing on the training time per batch. As shown in Table. 4, our method achieves the second-fastest training time, surpassed only by KDSNN. This advantage is largely due to KDSNN simplifying the original KL divergence into cross-entropy in the logit-based framework, while lowering computational cost, also affect accuracy. Our approach outperforms other techniques by utilizing only the logits output for knowledge distillation, thus eliminating the need for additional auxiliary training modules or feature map comparisons. In contrast, previous methods demand more time and resources to extract knowledge from intermediate layers.

Visualization. We employ t-SNE to visualize features learned from various distillation methods on the CIFAR100 dataset, utilizing ResNet-34 as the teacher and ResNet-18 as the student model. As shown in Fig. 5, our approach significantly enhances the distinguishability of deeper features compared to existing SNN distillation techniques.

5. Conclusion

This work addresses the issue of existing SNN distillation methods that are simply derived from ANN. We propose a new framework specifically designed for SNN distillation, which better exploits the temporal information contained in the logit output of SNNs for the process of knowledge distillation. Specifically, we incorporate a temporal separation strategy and introduce an entropy regularization term into the original distillation method. We aim to uncover the potential of SNN distillation by fully leveraging temporal information and rectifying the erroneous teacher knowledge. Extensive experimental results demonstrate the efficacy of this approach.

6. Acknowledgements

This work was supported in part by National Natural Science Foundation of China (NSFC) (62476035, 62206037, 62276230), and Natural Science Foundation of Zhejiang Province (LDT23F02023F02), and State Key Laboratory (SKL) of Biobased Transportation Fuel Technology.

References

- [1] Tong Bu, Wei Fang, Jianhao Ding, PengLin Dai, Zhaofei Yu, and Tiejun Huang. Optimal ann-snn conversion for high-accuracy and ultra-low-latency spiking neural networks. *arXiv preprint arXiv:2303.04347*, 2023. 1
- [2] Yongqiang Cao, Yang Chen, and Deepak Khosla. Spiking deep convolutional neural networks for energy-efficient object recognition. *International Journal of Computer Vision*, 113:54–66, 2015. 1
- [3] Fabio Cermelli, Massimiliano Mancini, Samuel Rota Buló, Elisa Ricci, and Barbara Caputo. Modeling the background for incremental learning in semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9233–9242, 2020. 3
- [4] Li Chen, Chunyan Yu, and Lvcai Chen. A new knowledge distillation for incremental object detection. In *2019 International Joint Conference on Neural Networks (IJCNN)*, pages 1–7. IEEE, 2019. 3
- [5] Jang Hyun Cho and Bharath Hariharan. On the efficacy of knowledge distillation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4794–4802, 2019. 2, 3
- [6] Mike Davies, Narayan Srinivasa, Tsung-Han Lin, Gautham Chinya, Yongqiang Cao, Sri Harsha Choday, Georgios Dimou, Prasad Joshi, Nabil Imam, Shweta Jain, et al. Loihi: A neuromorphic manycore processor with on-chip learning. *Ieee Micro*, 38(1):82–99, 2018. 1
- [7] Haoyu Deng, Ruijie Zhu, Xuerui Qiu, Yule Duan, Malu Zhang, and Liang-Jian Deng. Tensor decomposition based attention module for spiking neural networks. *Knowledge-Based Systems*, 295:111780, 2024. 6
- [8] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009. 2
- [9] Lei Deng, Yujie Wu, Xing Hu, Ling Liang, Yufei Ding, Guoqi Li, Guangshe Zhao, Peng Li, and Yuan Xie. Rethinking the performance comparison between snns and anns. *Neural networks*, 121:294–307, 2020. 1
- [10] Shikuang Deng and Shi Gu. Optimal conversion of conventional artificial neural networks to spiking neural networks. *arXiv preprint arXiv:2103.00476*, 2021. 1, 3
- [11] Shikuang Deng, Yuhang Li, Shanghang Zhang, and Shi Gu. Temporal efficient training of spiking neural network via gradient re-weighting. In *International Conference on Learning Representations*. 6, 7
- [12] Peter U Diehl, Daniel Neil, Jonathan Binas, Matthew Cook, Shih-Chii Liu, and Michael Pfeiffer. Fast-classifying, high-accuracy spiking deep networks through weight and threshold balancing. In *2015 International joint conference on neural networks (IJCNN)*, pages 1–8. ieee, 2015. 3
- [13] Jianhao Ding, Tong Bu, Zhaofei Yu, Tiejun Huang, and Jian Liu. Snn-rat: Robustness-enhanced spiking neural network through regularized adversarial training. *Advances in Neural Information Processing Systems*, 35:24780–24793, 2022. 1
- [14] Jason K Eshraghian, Max Ward, Emre O Neftci, Xinxin Wang, Gregor Lenz, Girish Dwivedi, Mohammed Ben-namoun, Doo Seok Jeong, and Wei D Lu. Training spiking neural networks using lessons from deep learning. *Proceedings of the IEEE*, 2023. 1
- [15] Wei Fang, Zhaofei Yu, Yanqi Chen, Tiejun Huang, Timothée Masquelier, and Yonghong Tian. Deep residual learning in spiking neural networks. *Advances in Neural Information Processing Systems*, 34:21056–21069, 2021. 3, 7
- [16] Isha Garg, Sayeed Shafayet Chowdhury, and Kaushik Roy. Dct-snn: Using dct to distribute spatial information over time for low-latency spiking neural networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4671–4680, 2021. 3
- [17] Yufei Guo, Weihang Peng, Yuanpei Chen, Liwen Zhang, Xi-aode Liu, Xuhui Huang, and Zhe Ma. Joint a-snn: Joint training of artificial and spiking neural networks via self-distillation and weight factorization. *Pattern Recognition*, 142:109639, 2023. 3
- [18] Yufei Guo, Yuhang Zhang, Yuanpei Chen, Weihang Peng, Xi-aode Liu, Liwen Zhang, Xuhui Huang, and Zhe Ma. Membrane potential batch normalization for spiking neural networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 19420–19430, 2023. 6, 7
- [19] Bing Han and Kaushik Roy. Deep spiking neural network: Energy efficiency through time based coding. In *European conference on computer vision*, pages 388–404. Springer, 2020. 3
- [20] Bing Han, Gopalakrishnan Srinivasan, and Kaushik Roy. Rmp-snn: Residual membrane potential neuron for enabling deeper high-accuracy and low-latency spiking neural network. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 13558–13567, 2020. 3
- [21] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 5
- [22] Geoffrey Hinton. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015. 3
- [23] Geoffrey Hinton, Li Deng, Dong Yu, George E Dahl, Abdelrahman Mohamed, Navdeep Jaitly, Andrew Senior, Vincent Vanhoucke, Patrick Nguyen, Tara N Sainath, et al. Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups. *IEEE Signal processing magazine*, 29(6):82–97, 2012. 1
- [24] Di Hong, Jiangrong Shen, Yu Qi, and Yueming Wang. Lasnn: Layer-wise ann-to-snn distillation for effective and efficient training in deep spiking neural networks. *arXiv preprint arXiv:2304.09101*, 2023. 2, 3, 6, 7
- [25] Saeed Reza Kheradpisheh, Mohammad Ganjtabesh, Simon J Thorpe, and Timothée Masquelier. Sdp-based spiking deep convolutional neural networks for object recognition. *Neural Networks*, 99:56–67, 2018. 2
- [26] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009. 2
- [27] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural net-